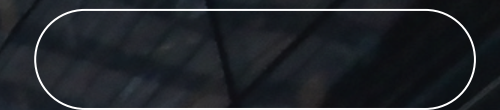




Discovering Emerging Topics in Drug Discovery Research Using BERTopic

Topic Modeling



Project Overview & Motivation

- Topic modeling on scientific publications in drug discovery
- Combines transformer embeddings + clustering
- Goal: To uncover hidden research themes and their evolution over time



Dataset & Preprocessing



Dataset

976 scientific abstracts

Fields used:

- Abstract text
- Publication year



Text preprocessing:

- Lowercasing
- Punctuation removal
- Stopword removal
- Lemmatization (spaCy)

- **Cleaned text is used for topic representation, not for embeddings**





Embeddings & Representation

- Sentence embeddings generated using Sentence-BERT
- Each abstract → 1 dense numerical vector
- Embedding shape: (976, 384)
- Captures semantic similarity, not keywords
- Allows abstracts discussing similar scientific ideas to be grouped together



```
import torch
torch.manual_seed(SEED)
model = SentenceTransformer('all-MiniLM-L6-v2')

embeddings = model.encode(
    df['clean_text'],
    show_progress_bar=True,
    convert_to_numpy=True
)
```



Topic Modeling with BERTopic

BERTopic pipeline:

- UMAP → dimensionality reduction
- HDBSCAN → density-based clustering
- c-TF-IDF → topic word representation

```
umap_model = UMAP(n_neighbors=10, n_components=5, min_dist=0.0, metric='cosine', random_state=42)
hdbscan_model = HDBSCAN(min_cluster_size=12, metric='euclidean', cluster_selection_method='eom', prediction_data=False)

topic_model = BERTopic(umap_model=umap_model, hdbscan_model=hdbscan_model, calculate_probabilities=False, # keeps mem
verbose=True)
```

Results:

- Identified 12 coherent topics
- Noise documents labeled as Topic -1

Major topics identified:

- Molecular docking & virtual screening
- Drug discovery method design
- Antiviral protease inhibitors
- Kinase inhibitors in cancer
- GPCR ligand–receptor studies

Topics are interpretable and domain-consistent

Zero-Shot Topic Labeling for Enhanced Interpretability

- Applied BART-MNLI zero-shot classification
- Automatically assigned semantic labels to topics
- No manual annotation required
- Improves interpretability of unsupervised results

```
candidate_labels = [  
    "molecular docking and virtual screening",  
    "antiviral drug discovery (protease inhibitors)",  
    "cancer therapeutics (kinase inhibitors, anticancer compounds)",  
    "neurodegenerative disease drug discovery (Alzheimer's)",  
    "antibiotic discovery and resistance (bacterial targets)",  
    "tuberculosis drug discovery",  
    "antimalarial drug discovery",  
    "GPCR drug discovery (ligands and receptors)",  
    "metabolic disease therapeutics (diabetes enzymes)",  
    "protein-protein interaction inhibitors",  
    "drug target identification and validation"  
]
```

```
zero_shot_labels  
  
{0: 'molecular docking and virtual screening',  
 1: 'drug target identification and validation',  
 2: 'molecular docking and virtual screening',  
 3: 'molecular docking and virtual screening',  
 4: 'drug target identification and validation',  
 5: 'molecular docking and virtual screening',  
 6: 'antimalarial drug discovery',  
 7: 'drug target identification and validation',  
 8: 'molecular docking and virtual screening',  
 9: 'tuberculosis drug discovery',  
10: 'molecular docking and virtual screening',  
11: 'drug target identification and validation'}  
  
info = topic_model.get_topic_info().copy()  
info["zero_shot_label"] = info["Topic"].map(zero_shot_labels)  
info
```

	Topic	Count	Name	Representation	Representative_Docs	zero_shot_label
0	-1	256	-1_compound_inhibitor_drug_study	[compound, inhibitor, drug, study, molecular, ...]	[background delta play key role bcell signal t...	NaN
1	0	260	0_docking_ligand_protein_bind	[docking, ligand, protein, bind, drug, screeni...	[computational docking means prioritise small ...]	molecular docking and virtual screening
2	1	108	1_drug_discovery_method_design	[drug, discovery, method, design, model, molec...	[drug designing development important area res...	drug target identification and validation
3	2	106	2_virus_drug_protease_antiviral	[virus, drug, protease, antiviral, viral, comp...	[emergence new variant raise concern effective...	molecular docking and virtual screening
4	3	47	3_kinase_inhibitor_compound_cell	[kinase, inhibitor, compound, cell, cancer, ta...	[kinase key target approximately breast cancer...	molecular docking and virtual screening
5	4	44	4_ad_disease_compound_inhibitor	[ad, disease, compound, inhibitor, alzheimer, ...]	[discovery novel multifunctional inhibitor tar...	drug target identification and validation





Topic Evolution Over Time & Conclusion

Topics analyzed across publication years

Observations:

- Growth in molecular docking research
- Increasing focus on antiviral and cancer targets
- antimalarial drug is more of a seasonal research area

Conclusion:

- BERTopic effectively reveals trends and emerging themes
- Useful tool for literature analysis in biomedical research

Topic		Words	Frequency	Timestamp
0	-1	inhibitor, compound, design, covalent, drug	14	2015
1	0	docking, protein, screening, discovery, virtual	29	2015
2	1	drug, discovery, designing, probe, anesthetic	4	2015
3	2	virus, dengue, hbv, pr, protease	3	2015
4	3	biphenyl, amide, inhibitor, indicate, value	1	2015
...	...	...	...	...
125	4	fisher, target, identification, aid, perresidue	1	2025
126	5	caffeic, phytoconstituent, ehretia, acid, energy	1	2025
127	8	kaempferol, alphaamylase, alphaglucosidase, an...	2	2025
128	9	tb, tuberculosis, katg, inha, inh	2	2025
129	10	pneumoniae, leprosy, murg, antibiotic, high	2	2025

13	3	antileishmanial, vl, endemic, leishmaniasis, b...	1	2017
14	0	drug, molecular, compound, target, discovery	89	2018
15	1	virus, influenza, antiviral, ipnv, polymerase	5	2018
16	2	ache, inhibitor, study, compound, value	4	2018
17	3	repellent, repellence, mmhg, pfatc, denovo	3	2018
18	0	drug, compound, molecular, docking, discovery	77	2019
19	1	antiviral, drug, virus, flavonoid, activity	3	2019
20	2	ad, inhibitor, bche, compound, disease	11	2019
21	3	new, antimalarial, vhts, brammt, target	1	2019
22	-1	rna, chemclipfragmap, small, structure, fragment	2	2020
23	0	drug, compound, molecular, discovery, docking	68	2020
24	1	drug, virus, protease, antiviral, zikv	20	2020
25	2	ad, ache, compound, inhibitor, disease	7	2020
26	3	worm, adult, schistosomula, mansonii, antimalarial	2	2020
27	0	drug, molecular, compound, target, docking	81	2021
28	1	drug, protease, compound, virus, molecular	27	2021
29	2	disease, neuropathic, pain, parkinson, neurode...	5	2021
30	0	drug, molecular, docking, target, compound	53	2022
31	1	drug, viral, inhibitor, virus, compound	14	2022
32	2	ad, drug, target, shortlist, specific	1	2022
33	3	antimalarial, drug, tbgpiplc, oxadiazole, para...	4	2022

33	3	antimalarial, drug, tbgpiplc, oxadiazole, para...	4	2022
34	-1	rna, chemistry, target, design, medicinal	1	2023
35	0	drug, docking, molecular, discovery, target	61	2023
36	1	protease, mpro, main, drug, study	6	2023
37	2	maob, work, inhibitor, sbvs, target	1	2023
38	3	drug, parasite, rpob, antimalarial, rifampicin	3	2023
39	0	drug, molecular, compound, discovery, target	156	2024
40	1	compound, drug, bind, potential, virus	13	2024
41	2	ad, disease, compound, abeta, aggregation	11	2024
42	3	antimalarial, drug, compound, malaria, molecule	6	2024

thank you



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